

# **Bangladesh Army University of Science and Technology**

## **Saidpur, Nilphamari**



## **Department Based study Planner**

### **Submitted By**

**Student Name**

**Student ID**

1. Rezwan Ahmed Torabee 0802510205101024

### **Submitted To**

**Teachers Name**

**Designation**

1. Md. Nahid Anam Opu Lecturer
2. Munira Akter Mou Lecturer

## **SESSIONAL PROJECT REPORT**

This Report is Presented in Partial Fulfillment of the course “**CSE 2100: Software Project I Sessional**” in the Department of Computer Science and Engineering

**June 7, 2026**

# DECLARATION

We hereby declare that this sessional project has been done by us under the supervision of **Md. Nahid Anam Opu, Lecturer & Munira Akter Mou, Lecturer** Department of Computer Science and Engineering (CSE), Bangladesh Army University of Science and Technology (BAUST). We also declare that neither this project nor any part of this project has been submitted elsewhere as sessional project.

## **Submitted To:**

### **Nahid Anam Opu**

Lecturer  
Department of Computer Science and  
Engineering  
Bangladesh Army University of Science and  
Technology

### **Munira Akter Mou**

Lecturer  
Department of Computer Science and  
Engineering  
Bangladesh Army University of Science and  
Technology

## **Submitted by**

Rezwan Ahmed Torabee  
0802510205101024  
Dept. of CSE, BAUST

# **DEPARTMENT BASED STUDY PLANNER PROJECT REPORT**

## **1. Abstract**

The Department-Based Study Planner is a JavaFX-based desktop application developed to help students organize and manage their academic tasks efficiently. Different academic departments require different study approaches and activities. Therefore, this application provides department-specific study suggestions based on the selected department.

The system allows students to register using their name, ID, and department information. After successful registration, students can access a personalized planner where they can add suggested tasks or create custom tasks. To encourage focused planning, the application limits the number of active tasks to five. Completed tasks can be removed easily using the task completion feature.

The project was developed using Java, JavaFX, Visual Studio Code, Git, and GitHub. The final system successfully demonstrates graphical user interface development, event-driven programming, input validation, task management, and Object-Oriented Programming concepts.

The project provides a simple, user-friendly, and effective solution for academic task organization while creating opportunities for future enhancements such as database integration, reminders, and cloud synchronization.

## **2. Team Members and Contributions**

### **Member 1: Rezwan Ahmed (0802510205101024)**

Requirement Analysis, JavaFX Development, System Planning, User Interface Design, Input Validation Implementation, Task Management Module Development, Testing and Debugging, Documentation, GitHub Repository Management

## **3. Introduction**

### **Problem Statement**

Students often struggle to manage academic responsibilities such as assignments, projects, examination preparation, and skill development activities. Most students rely on traditional note-taking methods or generic task managers that do not provide academic-specific guidance. As a result, students frequently experience poor organization, missed deadlines, and inefficient study routines.

The Department-Based Study Planner addresses these challenges by providing personalized study suggestions and task management capabilities based on a student's academic department.

## **Objectives**

- Develop a desktop-based study planner using JavaFX.
- Provide department-specific study recommendations.
- Implement student registration functionality.
- Validate user input information.
- Allow users to create and manage tasks.
- Implement task completion functionality.
- Demonstrate practical software development skills.

## **Scope**

The application focuses on academic task planning for students of different departments. The current version supports CSE, BBA, and English departments and provides department-specific study recommendations and task management functionality.

## **4. Planning and Analysis**

### **Project Overview**

The Department-Based Study Planner is a desktop application that assists students in organizing study activities through personalized task suggestions and task management features.

The application follows a simple workflow:

Student Registration → Validation → Planner Dashboard → Task Management

### **Functional Requirements**

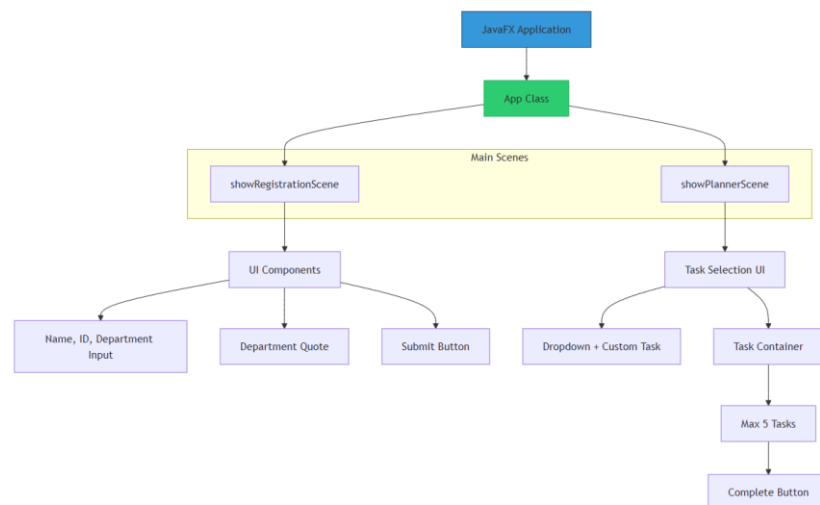
- Register student information.
- Validate student name and ID.
- Allow department selection.
- Display department-specific study suggestions.
- Allow custom task creation.

- Add tasks to a planner list.
- Remove completed tasks.
- Restrict task quantity to five active tasks.

## Non-Functional Requirements

- User-Friendly Interface
- Fast Response Time
- Reliability
- Simplicity
- Maintainability
- Portability
- Easy Navigation
- Efficient Resource Usage

## 5. System Design



## 6. Implementation

### Tools and Technologies Used

| Tool               | Purpose                                  |
|--------------------|------------------------------------------|
| Java 25            | Foundation runtime programming language. |
| JavaFX             | UI development, event handling,          |
| Visual Studio Code | Chosen IDE platforms.                    |
| Git                | Local version control and branching.     |
| GitHub             | Repository Hosting                       |

## Project structure

```
Department-Based-Study-Planner/
├── .github/
├── app/
│   ├── src/
│   │   ├── main/
│   │   │   ├── java/
│   │   │   │   ├── com/
│   │   │   │   │   ├── application/
│   │   │   │   │   │   App.java
│   │   │   │   │   └── module-info.java
│   │   │   └── resources/
│   │   │       ├── com/
│   │   │       │   ├── application/
│   │   │       │   │   ├── primary.fxml
│   │   │       │   │   ├── secondary.fxml
│   │   │       │   └── StudyPlanner.png
│   │   └── target/
│   │       ├── classes/
│   │       └── test-classes/
└── pom.xml
```

## Major Modules

### 1. Registration Module

- Name
- Student ID
- Department

Performs input validation before proceeding.

### 2. Department Suggestion Module

Provides department-specific study recommendations.

CSE:

- Practice Problem Solving
- Complete DSA Assignment
- Practice OOP Java
- Work on SDP Project
- Prepare for Exams

### 3. Task Management Module

- Add Task
- Custom Task Creation
- Complete Task
- Task Limit Management

## Code Snippets

```
private void addTaskRow(String taskName) {
    HBox row = new HBox();
    row.setAlignment(Pos.CENTER_LEFT);
    row.setPadding(new Insets(15, 20, 15, 20));
    row.setStyle(
        "-fx-background-color: white; -fx-border-color: #e9ecef; -fx-border-radius: 10;
        -fx-background-radius: 10;");

    Label taskLabel = new Label(taskName);
    taskLabel.setStyle("-fx-font-size: 16px; -fx-text-fill: #000000; -fx-font-weight: 500;");

    Region spacer = new Region();
    HBox.setHgrow(spacer, Priority.ALWAYS);

    Button completeBtn = new Button("Complete");
    completeBtn.setStyle(
        "-fx-background-color: #2ecc71; -fx-text-fill: white; -fx-font-weight: bold; -fx-font-size:
        14px; -fx-padding: 8 18; -fx-background-radius: 8;");

    completeBtn.setOnAction(e -> {
        taskContainer.getChildren().remove(row);
        limitErrorLabel.setText("");
    });
}
```

## 7. Testing

### Test case & results

| Test ID | Scenario Description      | Tested Inputs            | Expected Response                    | Verdict |
|---------|---------------------------|--------------------------|--------------------------------------|---------|
| TC-01   | Name input must be String | String & Integers        | Name must contain only letters       | Passed  |
| TC-02   | ID input must be integer  | Integers & String        | Name required & ID must be 3 digits! | Passed  |
| TC-03   | Can only add 5 tasks      | Add 6 <sup>th</sup> task | You can only add 5 tasks !           | Passed  |

## Testing Evidence

The image displays three screenshots of the 'Department Based Study Planner' application, illustrating the testing evidence for the tasks listed above.

**Screenshot 1: Student Registration**  
 The 'Name' field contains 'Prince07'. An error message is displayed: 'Error: Name must contain only letters (no numbers)!'. The 'ID' field contains '354'. The 'Department' dropdown is set to 'CSE'. A blue 'Submit' button is visible at the bottom.

**Screenshot 2: Student Registration (Valid Input)**  
 The 'Name' field contains 'Prince'. The 'ID' field contains 'Abc5'. The 'Department' dropdown is set to 'CSE'. A blue 'Submit' button is visible at the bottom. An error message is displayed: 'Error: Name required & ID must be 3 digits!'. A tip at the bottom reads: 'Start Small But Start Today. \*Tip: Build a habit of coding at least 1 hour daily!'

**Screenshot 3: Welcome, Paris**  
 The user is logged in as 'Paris' in the 'CSE' department. The 'Select Task or Write Task' dropdown is set to 'Prepare for exams'. A blue '+ Add Task' button is visible. Below it, a red message states: 'You can only add 5 tasks !'. A list of tasks is shown with 'Complete' buttons: 'Practice problem solving', 'Complete DSA Assignment', 'Practice OOP JAVA', 'Work on SDP Project', and 'Complete Homework'.

## 8. Deployment Timeline

|                                                                                                                                                     |                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Update App.java</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 632ebe3...3fef5c0 • 7 days ago                                     | <b>Fixed Testcase User input</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • e0c9b1c...741277d • 2 minutes ago                      |
| <b>Update App.java</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • cbacf2d...632ebe3 • 8 days ago                                     | <b>Revise README for department-based study planner</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • dd3e27e...e0c9b1c • 6 hours ago |
| <b>Add UI elements to the application main scene</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 59f16de...cbacf2d • 9 days ago       | <b>Fixed error labeling and scene ratio</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • b1940d9...dd3e27e • 7 hours ago             |
| <b>Add task button and limit error handling</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 8a586bb...59f16de • 9 days ago            | <b>Increase window height and update UI styles</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 13c1780...b1940d9 • yesterday        |
| <b>Implement task selection</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 1644970...8a586bb • 10 days ago                           | <b>Code updated</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • f8c0862...13c1780 • 3 days ago                                      |
| <b>Implement showRegistrationScene</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 7345c02...1644970 • 12 days ago                    | <b>Refactor UI and enhance task management features</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 823c198...f8c0862 • 4 days ago  |
| <b>Update</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • e0b0363...7345c02 • 13 days ago                                             | <b>Update README.md</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • eaeb93e...823c198 • 6 days ago                                  |
| <b>Update</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 492959f...e0b0363 • 13 days ago                                             |                                                                                                                                                   |
| <b>Implement student registration and task planner UI</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 96e76e4...492959f • 13 days ago |                                                                                                                                                   |
| <b>My Project Upload</b><br>rezwan-ahmed-l7 pushed 1 commit to <code>main</code> • 23625f9...96e76e4 • 13 days ago                                  |                                                                                                                                                   |
| <b>Initial commit</b><br>rezwan-ahmed-l7 created <code>main</code> • 23625f9 • 13 days ago                                                          |                                                                                                                                                   |

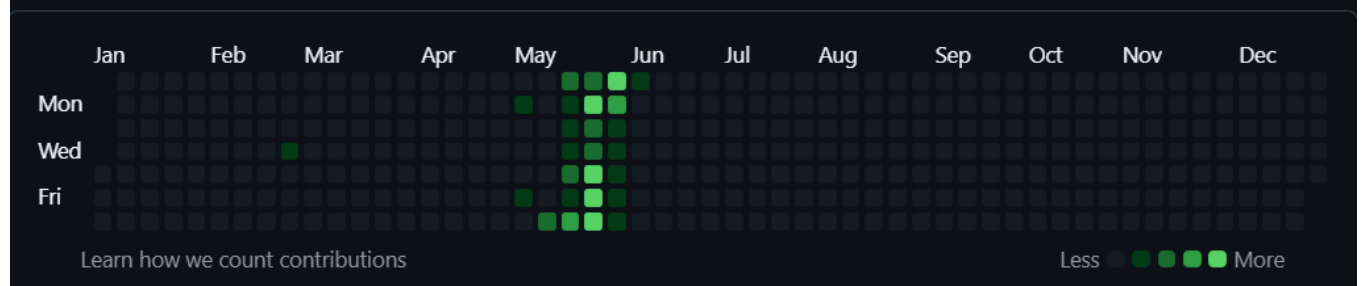
### Summary

Excluding merges, **1 author** has pushed **18 commits** to main and **18 commits** to all branches.

On main, **2 files** have changed and there have been **413 additions** and **46 deletions**

197 contributions in 2026

Contribution settings ▾



GitHub repo: <https://github.com/rezwan-ahmed-l7/Study-Planner-App>

## 9. Future Work

- Back button
- System Scalability Expansion
- UI Enhancement
- Data Persistence
- Calendar integration
- Timer integration